3 Adaptive Cruise Control System with platooning

A platoon of 5 vehicles is shown below [7]:

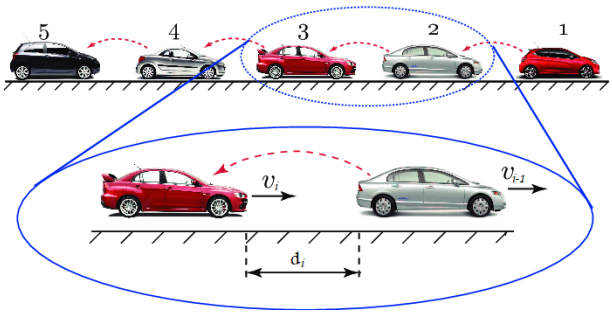

Figure 3: Car platoon with adaptive cruise control

3.1 System Modeling

$$\begin{cases} \dot{d}_i(t) = v_{i-1}(t) - v_i(t) \\ m_i \dot{v}_i(t) = u_i(t) - F_{\mathrm{L}i}(v_i(t)) \end{cases}, \quad i = 2, \dots, 5, \tag{7}$$

where d_i is the inter-vehicle distance, the 1st vehicle is the platoon leader, and v_i is the ith vehicle velocity. Moreover, m_i is the vehicle mass and $F_{\mathrm{L}i}(v_i)$ represents the aerodynamic drag and the rolling resistance forces, described as $F_{\mathrm{L}i}(v_i(t)) = \frac{1}{2}\rho c_{\mathrm{a}} A v_i^2(t) + c_{\mathrm{r}} m_i g_{\mathrm{r}}$, with ρ being the air density, c_{a} the aerodynamic coefficient, A is the equivalent vehicle surface, while c_{r} and g_{r} are the rolling resistance coefficient and the gravitational acceleration, respectively. As for the input force $u_i(t)$, it can be seen as the result of two additive components, representing the traction force component, positive during acceleration and negative during deceleration (regenerative braking), and the mechanical braking force. Such input has to be constrained between two values depending on the specific properties of tires and on conditions of the road, i.e., $u_i(t) \in [u_{\min}, u_{\max}]$ for all $i = 2, \dots, 5$, with $u_{\min} < 0$ and $u_{\max} > 0$ being the force bounds.

The desired distance among the vehicles is selected according to the Constant Time Headway (CTH) principle requiring the spacing among the cars to be defined as $d_i^{\mathrm{CTH}} = d_{\min} + t_{\mathrm{CTH}} v_i(t)$, for all $i = 2, \dots, 5$.

3.2 Control Objectives

The control objective is to regulate the vehicles velocity to a desired reference $v^\star(t)$, that is to take each $g_i(t, v) = v_i(t) - v^\star(t)$ to zero while controlling the inter-vehicle distance to the reference d_i^{CTH}.

Remark: it is supposed that $v^\star(t) = v_1(t)$.

3.3 Control Structure

For this application, use a single MPC controller with:

- Four manipulated variable: vehicles traction force u_i, $i = 2, \dots, 5$.
- Eight measured outputs: vehicle position d_i and velocity u_i, $i = 2, \dots, 5$.
- One unmeasured disturbance: leading car velocity v_1.

3.4 Parameters values & Constraints

The initial velocities of the five cars are chosen as $[10\ ,12\ ,15\ ,9\ 10]^\top$ m/s, while the initial distances are $[110\ ,130\ ,100\ ,160]^\top$ m.

Vehicles parameters:

Parameter	**Value**		**Description**
g_r	9.81	$[m/s^2]$	Gravitational constant
$c_a\,A$	0.88	$[ms^2]$	aerodynamic coefficient $\times$ vehicle surface
$d_{\min}$	4	$[m]$	Minimum inter-vehicle distance
ρ	1.22	$[kg/m^3]$	Air density
t_{CTH}	3	$[s]$	Time headway
m_1	790	$[kg]$	Mass of vehicle 1
m_2	820	$[kg]$	Mass of vehicle 2
m_3	760	$[kg]$	Mass of vehicle 3
m_4	860	$[kg]$	Mass of vehicle 4
m_5	800	$[kg]$	Mass of vehicle 5

Table 3: Vehicles parameters

Constraints:

- Safety distance constraint: $d_i \geq d_{\min}$
- Limited traction force:$-600 \leq u_i \leq 600$
- Limited traction force rate:$-20 \leq \Delta u_i \leq 20$